s.no	Lanthanoids	Actinoids
5	They do not form oxo cations	They do form oxo cations such as UO_2^{2+}, NpO_2^{2+} etc
6	Besides +3 oxidation states lanthanoids show +2 and +4 oxidation states in few cases.	Besides +3 oxidation states actinoids show higher oxidation states such as +4, +5, +6 and +7.

Summary

- IUPAC defines transition metal as an element whose atom has an incomplete d sub shell or which can give rise to cations with an incomplete d sub shell. They occupy the central position of the periodic table, between s and p block elements,
- d- Block elements composed of 3d series (4th period) Scandium to Zinc (10 elements), 4d series (5th period) Yttrium to Cadmium (10 elements) and 5d series (6th period) Lanthanum, Haffinium to mercury.
- the general electronic configuration of d- block elements can be written as [Noble gas] $(n-1)d^{1-10}ns^{1-2}$, Here, n = 4 to 7 . In periods 6 and 7, the configuration includes $((n-2)f$ orbital ; [Noble gas] $(n-2)f^{14}(n-1)d^{1-10}ns^{1-2}$.
- All the transition elements are metals. Similar to all metals the transition metals are good conductors of heat and electricity. Unlike the metals of Group-1 and group-2, all the transition metals except group 11 elements are hard.
- As we move from left to right along the transition metal series, melting point first increases as the number of unpaired d electrons available for metallic bonding increases, reach a maximum value and then decreases, as the d electrons pair up and become less available for bonding.
- Ionization energy of transition element is intermediate between those of s and p block elements. As we move from left to right in a transition metal series, the ionization enthalpy increases as expected.
- The first transition metal Scandium exhibits only +3 oxidation state, but all other transition elements exhibit variable oxidation states by loosing electrons from (n-1)d orbital and ns orbital as the energy difference between them is very small.
- In 3d series as we move from Ti to Zn, the standard reduction potential $\left(E^0_{M^{2+}/M}\right)$ value is approaching towards less negative value and copper has a positive reduction potential. i.e., elemental copper is more stable than Cu^{2+}.

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